



# LINEAR ALGEBRA

**UE19MA251**

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## Tests for Positive definiteness

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## Agenda

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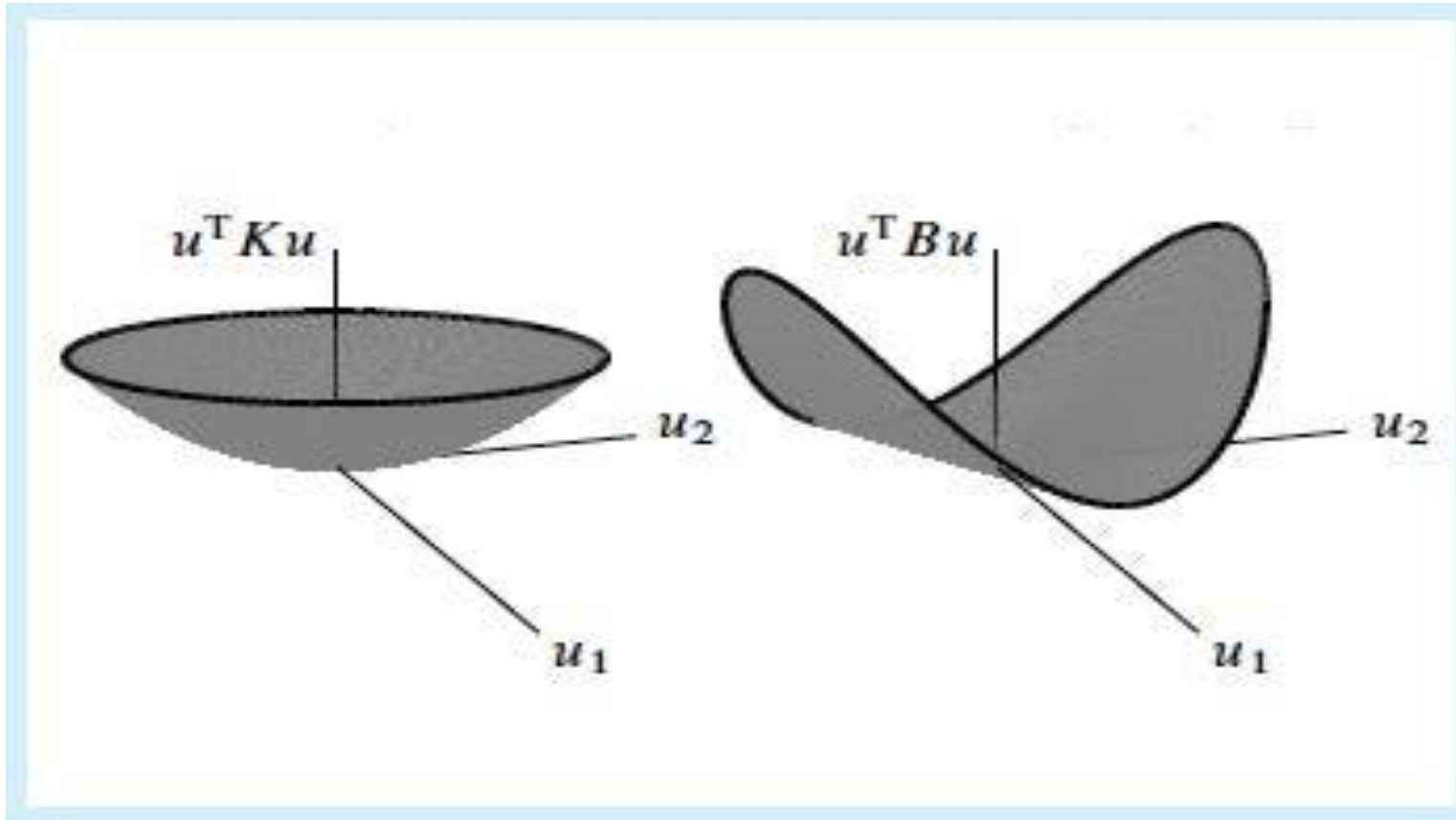


- Quadratic form - Classification
- Positive Definite Matrix –  $A+B$ ,  $A^T A$
- Positivity of Eigenvalues
- Equivalent statements for positive definiteness
- Examples

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## Quadratic Form: Classification

Positive definite and semidefinite: graphs of  $x^T A x$ .



<https://ocw.mit.edu/courses/mathematics/18-06sc-linear-algebra-fall-2011/positive-definite-matrices-and-applications/>

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## Quadratic Form: Classification

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Based on the sign of the quadratic form,  
SVD can be classified into 3 categories:

Positive definite if (Quadratic form)  $> 0$

Positive semi definite if (Quadratic form)  $\geq 0$

Negative definite if (Quadratic form)  $< 0$

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$A^T A$  is Positive Definite

$A^T A$  is symmetric and square.

Quadratic form:  $x^T A^T A x = (Ax)^T (Ax) = \|Ax\|^2 > 0$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} ; \quad A^T A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 10 & -1 \\ -1 & 5 \end{bmatrix}$$

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## Positive Definiteness Of $A+B$

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If  $A$  and  $B$  are positive definite, then so is  $A+B$ .

$$x^T(A+B)x = x^T A x + x^T B x > 0$$

$$\because x^T A x > 0 \text{ and } x^T B x > 0 .$$

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## POSITIVITY OF EIGENVALUES

Every eigenvalue of a positive definite matrix is positive.

**Proof.** Suppose  $\mathbf{A}$  is a positive definite matrix. Let  $\lambda$  be an eigenvalue of  $\mathbf{A}$ , and  $\mathbf{s}$  be an eigenvector of  $\mathbf{A}$  corresponding to  $\lambda$ . We have

$$\mathbf{A}\mathbf{s} = \lambda\mathbf{s}$$

It follows that

$$\mathbf{s}^T \mathbf{A} \mathbf{s} = \lambda(\mathbf{s}^T \mathbf{s})$$

Hence

$$\lambda = \frac{\mathbf{s}^T \mathbf{A} \mathbf{s}}{\mathbf{s}^T \mathbf{s}} > 0$$

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## POSITIVITY OF EIGENVALUES

A matrix is positive definite if every eigenvalue of the matrix is positive.

**Proof.** Suppose every eigenvalue of  $A$  is positive. By spectral theorem,  $A$  has an eigenvalue decomposition  $A = Q\Lambda Q^T$ . It follows that

$$x^T A x = \overbrace{x^T Q}^{y^T} \Lambda \overbrace{Q^T x}^y = y^T \Lambda y = \sum_i \lambda_i y_i^2$$

Hence, the quadratic form  $x^T A x$  is positive for any  $x \neq 0$ , and  $A$  is positive definite.

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## Equivalent statements for Positive definiteness

There are many ways to say a matrix is positive definite.

- ①  $A$  is positive definite.
- ② Every eigenvalue of  $A$  is positive.
- ③ The determinant of every leading principal sub-matrices of  $A$  is positive.
- ④  $A$  has full positive pivots.

What we have shown in the previous slides are

$$\textcircled{1} \Leftrightarrow \textcircled{2}$$

and

$$\textcircled{1} \Rightarrow \textcircled{3} \Rightarrow \textcircled{4} \Rightarrow \textcircled{1}$$

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## Examples



$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \quad \text{The quadratic form of } A \text{ is}$$

The quadratic form of  $A$  is

$$\begin{aligned} \mathbf{x}^T A \mathbf{x} &= 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3 \\ &= 2 \left( x_1 - \frac{1}{2}x_2 \right)^2 + \frac{3}{2} \left( x_2 - \frac{2}{3}x_3 \right)^2 + \frac{4}{3}x_3^2 \end{aligned}$$

The eigenvalues, the determinants, and the pivots are

$$\text{spectrum}(A) = \{2, 2 \pm \sqrt{2}\}, \quad |A_1| = 2, \quad |A_2| = 3, \quad |A_3| = 4$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 2 & & \\ & \frac{3}{2} & \\ & & \frac{4}{3} \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & 1 & -\frac{2}{3} \\ 0 & 0 & 1 \end{bmatrix}$$

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## EXAMPLE

For what numbers 'b' is the following matrix positive semi-definite?

$$A = \begin{bmatrix} 2 & -1 & b \\ -1 & 2 & -1 \\ b & -1 & 2 \end{bmatrix}.$$

$$\text{Solution: } \begin{vmatrix} 2 & -1 & b \\ -1 & 2 & -1 \\ b & -1 & 2 \end{vmatrix} = -2b^2 + 2b + 4 \geq 0$$

$$\Rightarrow b^2 - b - 2 \leq 0$$

$$(b-1)(b-2) \leq 0 \quad -1 \leq b \leq 2$$



# THANK YOU

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